

14 MAY 2025

SPRINT-1

Data4HD

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Meeting Agenda



Recap of Tasks Assigned

Reviewing previously assigned tasks



Tasks Completed

Discussing tasks that have been finished



Detailed Look at Work Done

Examining the specifics of completed work



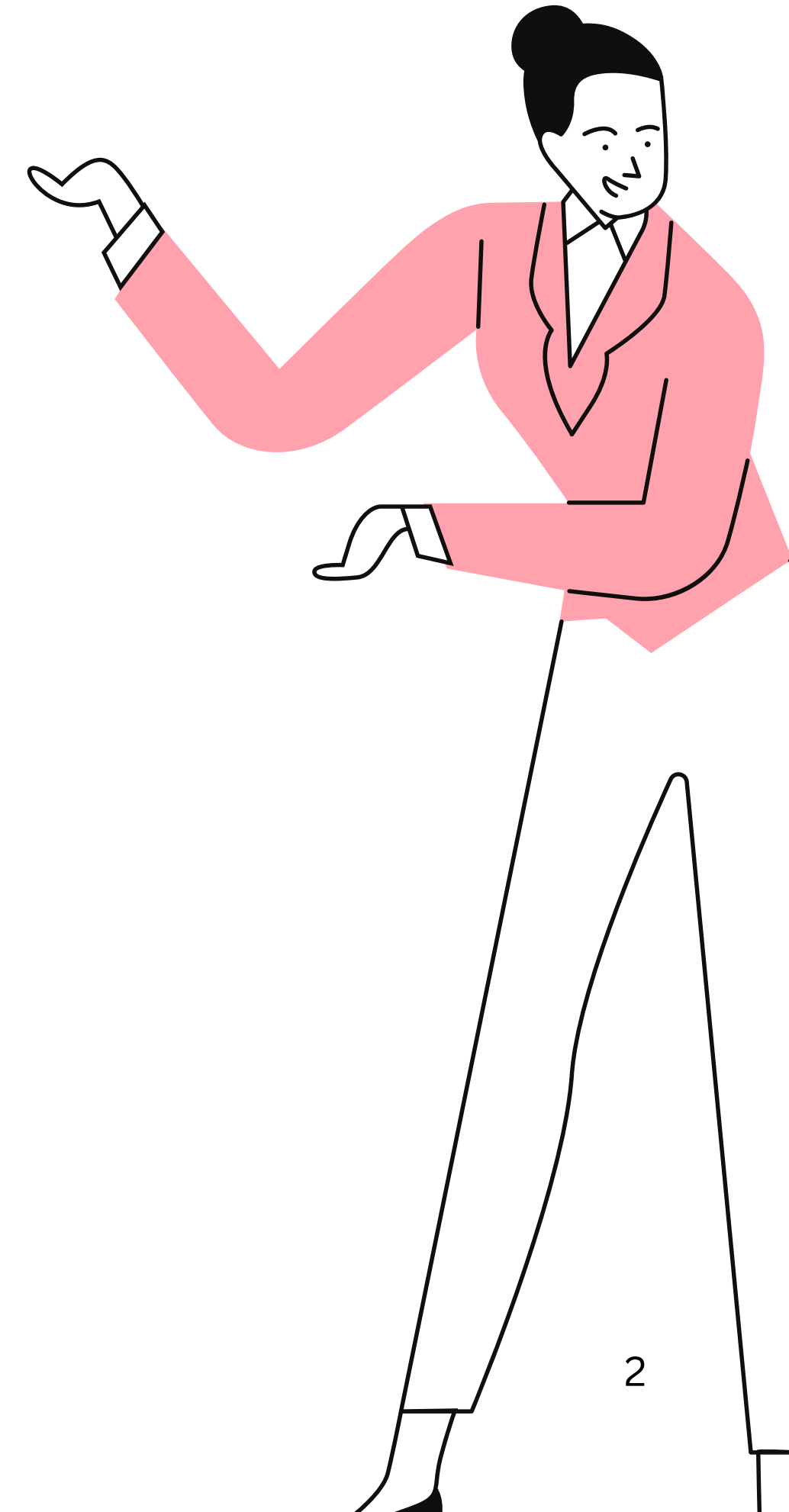
Key Data Issues Summary

Highlighting significant data-related problems



Q&A

Addressing questions and concerns



Tasks Assigned

1	Review Parking Data Examine Parking and Parken datasets
2	Identify Data Issues Find problems within the datasets
3	Develop User Stories Create narratives for user needs
4	Explore Open Data Search for additional data sources

Tasks Completed

City Dataset Exploration

Reviewed open data portals from other cities and documented it.



Dataset Review Parking & Parken

Familiarizing with the structure and content of datasets



User Story Development

Created user stories from Data Scientist's & Common User's perspective.

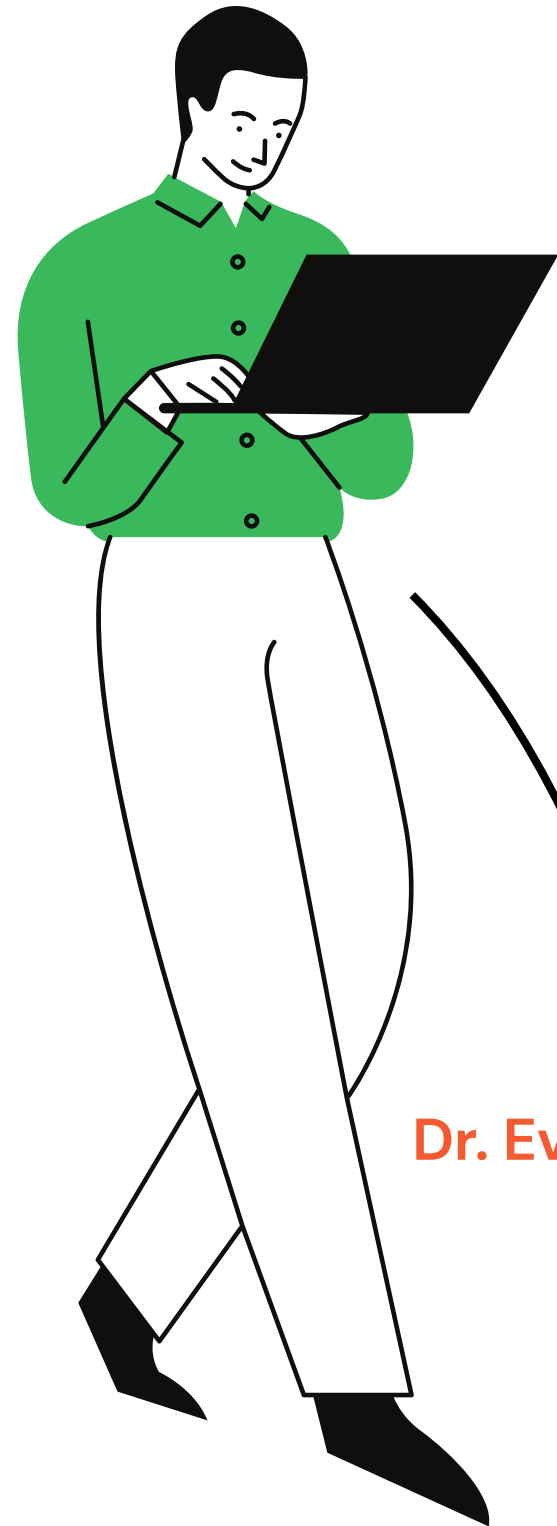


Data Exploration

Conducted EDA and identifying data quality issues



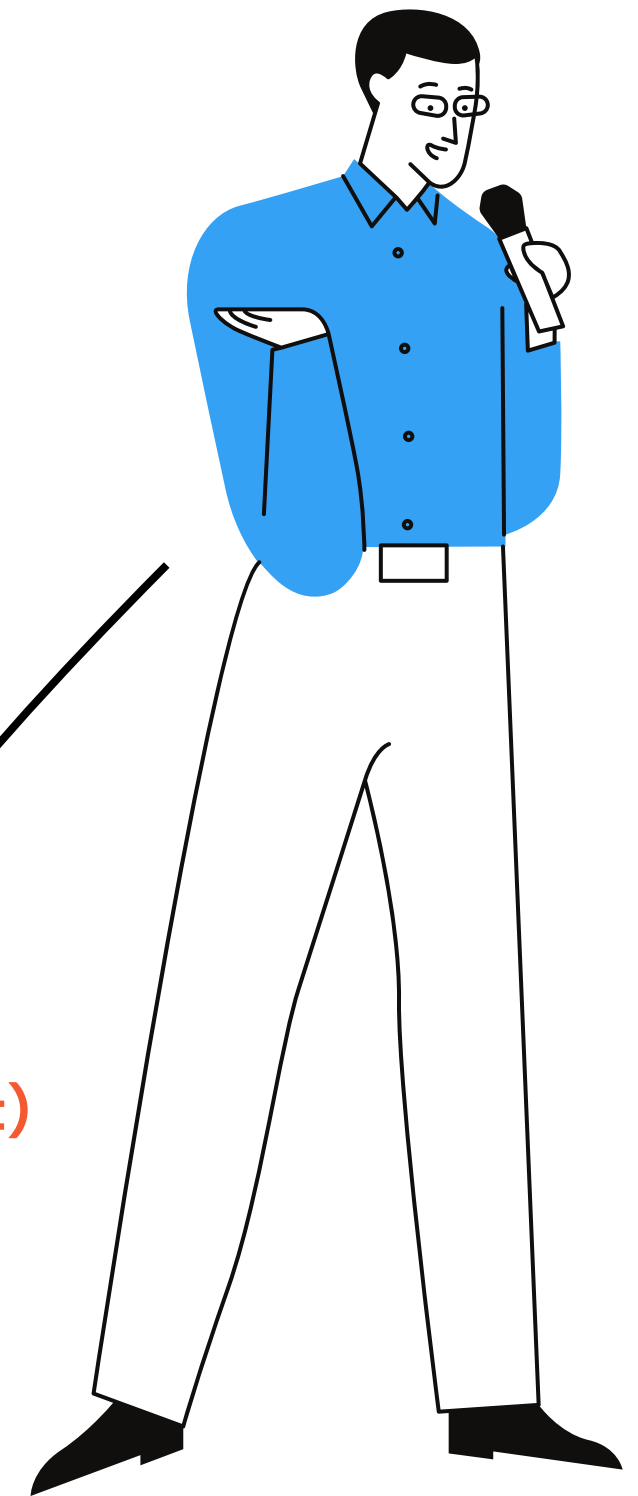
User Stories Developed



Dr. Evelyn Reed (Data Scientist)

Explored how different users engage with city data.

Read Full Personas



David Miller (Resident)

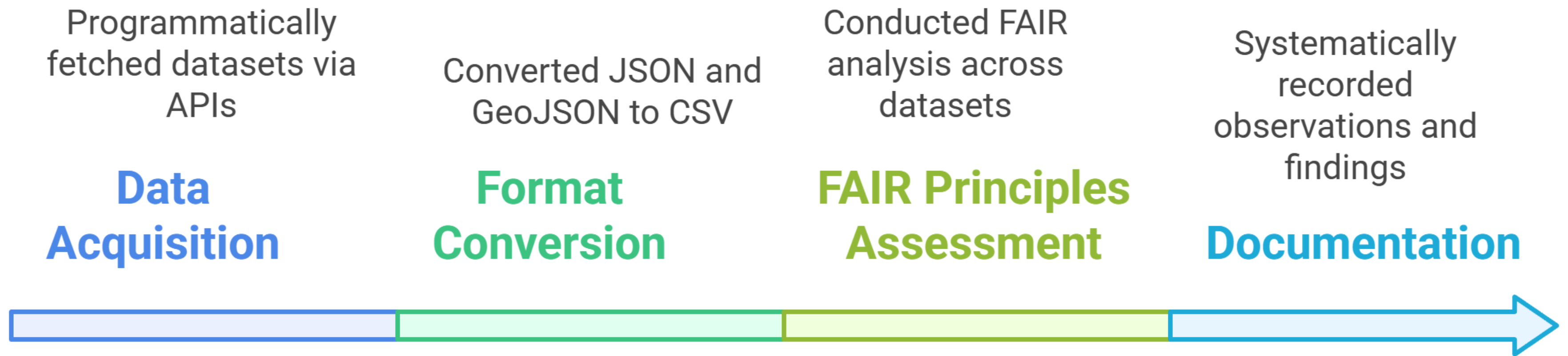
Exploration of Other City Open Data

Key Takeaways

- Best Overall: Bonn (modern, mobile-friendly, strong API, good updates).
- Best Multilingual: Open.NRW (consistent German + English support).
- Biggest Weakness: Cologne (dated, German-only, slow loading).
- Metadata Issues: All portals struggle with incomplete metadata (especially Hamburg/Cologne).
- Mobile Gaps: Hamburg and Cologne perform poorly on mobile.

Document Link

Parking & Parken Data – Overview



1) Parking Data Overview 2) Parken Data Overview

Data Issues Found : Disabled Parking Dataset

id	Notiz	BEZEICHNUN	BETREIBER	TYP	XTRID	URN	BESCHREIBU	BESCHRIFTU	PARKPLATZ_	
01e10d390a735e5b177c7a342de393dc	Ab, mit Basis gefunden	Kirchstraße 20-22	Stadt Heidelberg	öffentlich	45872814_3	F3E			Behinderte	
043c829a07b03ebccb4478b0f31b10c9		Mönchgasse / Heiliggeiststraße	Stadt Heidelberg	öffentlich	33077310_1				Behinderte	
053a58717092d3d800216e8d8601d6f6		Steubenstr. (Grahampark)/Doss. Landstr.	Stadt Heidelberg	öffentlich	33077229_2	E9A			Behinderte	
0586da1bb2270fe43e1425924b718137		Poststraße 17	Stadt Heidelberg	öffentlich	37638360_1				Behinderte	
075cf1e07c7c4d94ad838c2f29ceafb6		Blumenthalstraße 38	Stadt Heidelberg	öffentlich	33077358_1	F33			Mo-Fr 8-17	Behinderte
08a8109806edae3ee02a586466664d83		Untere Neckarstraße (ggü. Nr. 62)	Stadt Heidelberg	öffentlich	37638346_1	F26				Behinderte
09ccc50a00536c02677d8896aaa5d646	Baustelle	Blumenstraße 38	Stadt Heidelberg	öffentlich	46911969_1			Mo-Fr 8-17	Behinderte	
0b96b3c6e44e29b7715f3d97a785be5a	Baustelle	Blumenstraße 38	Stadt Heidelberg	öffentlich	46911969_4			Mo-Fr 8-17	Behinderte	
0bbdc0b10164013b61fc2139745fc479	Schild abgedeckt	Hardtstraße 2 (Zentralbetriebshof)	Stadt Heidelberg	öffentlich	33077379_5			beschränkt	Behinderte	
0bd248b45ff6ff516583cb173d8c08d7	Baustelle	Blumenstraße 38	Stadt Heidelberg	öffentlich	46911969_6			Mo-Fr 8-17	Behinderte	
0bec49d871a9a3d7bc467f5be9ce574c		Rohrbacher Straße (Bergfriedhof)	Stadt Heidelberg	öffentlich	33076962_2			7-19, 2h	Behinderte	
0c263c6a23b175d1fd5220ac9c88bc10		In der Neckarhelle 82	Stadt Heidelberg	öffentlich	33077136_1	F0C		Mo-Fr 8-18	Behinderte	
0c998587f851364b160967ef6ba5f157		Uferstraße gegenüber Nr. 38	Stadt Heidelberg	öffentlich	33076857_1	E7D			Behinderte	
106880d6e6c44d8453da8000f5e51503		Vangerowstraße 4 (Thermalbad)	Stadt Heidelberg	öffentlich	33076896_2	E7B			Behinderte	
11ba4dd3fa12cbd1d79d1e42d31985c8		Poststraße 5	Stadt Heidelberg	öffentlich	37638363_1				Behinderte	
1225ae5767942d51b3134746d85783ce		Kriegsstraße 16	Stadt Heidelberg	öffentlich	33077316_1			8-19, 2h	Behinderte	
12bf420df071badb2666b5b1dc080073	Keine Klare Markierung	Tiergartenstraße (Sportzentrum Nord)	Stadt Heidelberg	öffentlich	33076986_1				Behinderte	
1300d6ad34bdb1eb752a06158a545733		Dossenheimer Landstraße 18a	Stadt Heidelberg	öffentlich	37638369_1	F30		beschränkt	Behinderte	
1424c3892cf5a7adb2b01b01cbf6ea8b		Bergheimer Straße 63	Stadt Heidelberg	öffentlich	45872823_1	F31		Mo-Fr 8-19	Behinderte	
14e59663f615ede93361b26604822f09		Akademiestraße	Stadt Heidelberg	öffentlich	33076929_3				Behinderte	
14e9d305536da7c70ce947aa099e9c19	Weg	Pleikartsförster Straße (Sportz. Süd)	Stadt Heidelberg	öffentlich	33076959_2				Behinderte	
153dad0ac58f0ab11d81672d3088b5b0	Fläche uneben	Bahnhof Kirchheim / Hardtstraße	Stadt Heidelberg	öffentlich	45872817_2				Behinderte	
17ed0c6825cb14d0a334d295a9a70958		Friedhof Wieblingen	Stadt Heidelberg	öffentlich	33077010_1	F28		7-19, 2h	Behinderte	
1935904e1fc1c267a5c688126ce10de2		Bahnhofstraße 36	Stadt Heidelberg	öffentlich	45872820_1	F22			Behinderte	
193906de382f3f390d828fb9cb0722f7		Tiergartenstraße (Tiergartenschwimmbad)	Stadt Heidelberg	öffentlich	33077475_5				Behinderte	
1a77a4df7afd17c666ca4c0ed8fb69c1	Boden uneben	Hauptbahnhof (Willy-Brandt-Platz)	Stadt Heidelberg	öffentlich	33077007_5				Behinderte	
1c799d4dc8d260ca208274626ebc0450	Zu wenig/Keine Klare Mark.	Hofmeisterweg (Medizinische Klinik)	UniversitätsKlinikum	öffentlich	37638373_6				Behinderte	
1debcb7fdb2f187c97f162919885c5ec		Dantestraße 1	Stadt Heidelberg	öffentlich	33077040_1	F44		Mo-Fr 7-20	Behinderte	
1ed94a53094923b8760fed5cf19255cf	Zu wenig/Keine Klare Mark.	Hofmeisterweg (Medizinische Klinik)	UniversitätsKlinikum	öffentlich	37638373_7				Behinderte	
1ef1f38068fbcc481586f98f3c616b0a	Boden uneben	Hauptbahnhof (Willy-Brandt-Platz)	Stadt Heidelberg	öffentlich	33077007_4				Behinderte	
2134bb78881457a09c020d590b38bf96	Gibt es nicht?/Baustelle	Blunschlistraße 14	Stadt Heidelberg	öffentlich	33077415_1			Mo-Fr 8-12	Behinderte	
21dc48239ceb1f6a2483c3de2edb5489		Rohrbacher Straße 81	Stadt Heidelberg	öffentlich	33076905_1			beschränkt	Behinderte	
2205f9d0971f0cc0911d925415e21a8f		Heidelberger Straße 14	Stadt Heidelberg	öffentlich	45872835_1	F13		wtg8-19,2h	Behinderte	
22d502ff3b45bc1e529d4bdd015034dd		Im Emmertsgrund/Mombertplatz 33	Stadt Heidelberg	öffentlich	42214252_2				Behinderte	

Data Issues Found : Off Street Parking

_id	recvTime	fiwareServ	entityId	entityType	name	observatio	occupiedS	parkingSit	status	totalSpotNumber
1	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-0	33	20	0	672
2	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
3	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
4	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
5	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-0	33	20	0	672
6	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
7	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-0	33	20	0	672
8	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
9	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
10	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
11	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
12	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
13	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
14	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
15	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
16	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	39	20	0	672
17	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	40	20	0	672
18	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	40	20	0	672
19	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	40	20	0	672
20	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	40	20	0	672
21	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	40	20	0	672
22	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P20 HiP	2023-07-1	40	20	0	672

Data Issues Found : Off Street Parking

_id	recvTime	fiwareServ	entityId	entityType	name	observatio	occupiedS	parkingSiteId	status	totalSpotNumber	
1	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	74	19	0	353	
2	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	74	19	0	353	
3	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	74	19	0	353	
4	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	72	19	0	353	
5	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	73	19	0	353	
6	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	73	19	0	353	
7	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	71	19	0	353	
8	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	71	19	0	353	
9	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	68	19	0	353	
10	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	69	19	0	353	
11	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	66	19	0	353	
12	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	67	19	0	353	
13	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	61	19	0	353	
14	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	62	19	0	353	
15	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	57	19	0	353	
16	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	57	19	0	353	
17	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	56	19	0	353	
18	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	57	19	0	353	
19	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	56	19	0	353	
20	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	69	19	0	353	10
21	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	67	19	0	353	
22	2023-07-1	/main/par	urn:ngsiv2	OffStreetP	P19 Camp	2023-07-1	67	19	0	353	

Data Issues Found : Off Street Parking

entityId	entityType	name	observationDateTime	occupiedSpotNumber	parkingSiteId	status	totalSpotNumber	trend
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:05:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:10:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:15:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:20:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:25:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:35:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:45:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:50:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:30:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T15:00:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T15:10:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T16:15:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T16:55:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T17:10:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T15:35:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T15:55:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T16:00:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T16:20:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T16:45:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T16:40:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T17:15:00	0	26	Offen	55	0
urn:ngsiv2:OffStreetParking:P26	OffStreetParking	P26 Kirchheim	2023-07-05T14:55:00	0	26	Offen	55	0

Data Issues Found : Historical Data

areaServed	availableSpotNumber	description	latitude	longitude	maximumAllowedHeight	maximumAllowedWidth	name	observationDateTime	occupiedSpotNumber	outOfServiceSlotNumber	status	totalAvailableSpotNumber	totalSpotNumber
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:00:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:01:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:02:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:04:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:07:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:03:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:05:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:06:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:08:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:09:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:10:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:11:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:12:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:13:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:14:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:15:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:17:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:17:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:18:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:19:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:20:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:21:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:22:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:23:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:24:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:25:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:26:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:27:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:28:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:30:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:29:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:31:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:32:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:33:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:34:00	0	-55	spacesAvailable	55	0
	55	constant	49.376299	8.6585952	0	0	P26 Kirchheim	2023-03-23T15:37:00	0	-55	spacesAvailable	55	0

Data Issues Found : Historical Data

aggregateRati▼	c▼	areaServ▼	availableSpotNumb▼	descripti▼	latitude▼	longitu▼	maximumAllowedHeig▼	maximumAllowedWid▼	name▼	observationDateTin▼	occupiedSpotNumb▼	outOfServiceSlotNumb▼	stat▼	t
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:28:57	18			
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:28:57	18			
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:31:07	18			
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:33:15	18			
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:35:21	18			
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:37:27	18			
2.678571429	HIP?	HIP?	654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:39:32	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:41:34	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:43:37	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:45:41	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:47:44	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:49:46	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:51:52	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:54:02	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:56:05	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T11:58:08	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:00:14	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:02:17	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:04:20	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:06:25	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:08:31	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:10:36	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:12:40	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:14:46	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:16:48	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:18:53	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:20:54	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:22:58	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:24:59	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:25:00	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:27:08	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:29:13	18			
2.678571429	HIP?		654	constant	49.41068	8.68394			P20 HIP?	2023-04-12T12:31:16	18			

Data Issues Found : Off Street Parking

	recvTime	fiwareServicePath	entityId	entityType	areaServed	maximumAllowedWidth	latitude	description
1	2022-10-28T10:11:31.197Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	constant
2	2022-10-28T10:12:31.59Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	constant
3	2022-10-28T10:13:31.269Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
4	2022-10-28T10:14:31.62Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
5	2022-10-28T10:15:31.492Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
6	2022-10-28T10:16:31.194Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
7	2022-10-28T10:17:30.875Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
8	2022-10-28T10:18:31.74Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	increasing
9	2022-10-28T10:19:31.295Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	increasing
10	2022-10-28T10:20:30.795Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	increasing
11	2022-10-28T10:21:30.940Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	constant
12	2022-10-28T10:22:30.777Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
13	2022-10-28T10:23:31.371Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
14	2022-10-28T10:24:31.373Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
15	2022-10-28T10:25:31.491Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	constant
16	2022-10-28T10:26:31.439Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
17	2022-10-28T10:27:31.83Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
18	2022-10-28T10:28:30.629Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
19	2022-10-28T10:29:30.594Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
20	2022-10-28T10:30:31.423Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	constant
21	2022-10-28T10:31:31.207Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
22	2022-10-28T10:32:31.329Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
23	2022-10-28T10:33:31.304Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
24	2022-10-28T10:34:31.37Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
25	2022-10-28T10:35:30.598Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
26	2022-10-28T10:36:30.707Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
27	2022-10-28T10:37:31.171Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing
28	2022-10-28T10:38:31.177Z	/main/parking	urn:ngsiv2:OffStreetParking:P016	OffStreetParking	Zentrum	0	49.41364651	decreasing

Data Issues Found : Historical Data

	areaServed	availableSpotNumber	description	latitude	longitude	maximumAllowedHeight	maximumAllowedWidth	name	observationDateTime	occupiedSpotNumber	outOfServiceSlotNumber	status
cks	Campbell Barracks	83	increasing					P19 Campbell Barracks	2023-04-05T13:11:55	270		
cks	Campbell Barracks	83	constant					P19 Campbell Barracks	2023-04-05T13:13:59	270		
cks	Campbell Barracks	83	constant					P19 Campbell Barracks	2023-04-05T13:16:02	270		
cks	Campbell Barracks	83	constant					P19 Campbell Barracks	2023-04-05T13:18:05	270		
cks	Campbell Barracks	85	increasing					P19 Campbell Barracks	2023-04-05T13:20:08	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:22:11	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:24:17	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:26:19	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:28:23	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:30:26	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:32:34	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:34:39	268		
cks	Campbell Barracks	85	constant					P19 Campbell Barracks	2023-04-05T13:36:42	268		
cks	Campbell Barracks	86	increasing					P19 Campbell Barracks	2023-04-05T13:38:46	267		
cks	Campbell Barracks	87	increasing					P19 Campbell Barracks	2023-04-05T13:40:49	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:42:51	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:44:54	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:46:56	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:48:59	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:48:59	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:51:01	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:53:05	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:55:09	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:57:11	266		
cks	Campbell Barracks	87	constant					P19 Campbell Barracks	2023-04-05T13:59:16	266		
cks	Campbell Barracks	88	increasing					P19 Campbell Barracks	2023-04-05T14:01:18	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:03:20	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:05:24	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:07:27	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:09:34	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:11:39	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:13:43	265		
cks	Campbell Barracks	88	constant					P19 Campbell Barracks	2023-04-05T14:15:45	265		

Data Issues Found : Historical Data

observationDateTime	totalAvailableSpotNumber	occupiedSpotNumber	availableSpotNumber	name	maximumAllowedHeight	totalSpotNumber	aggregateRating	outOfServiceSlotNumber
2022-10-28T10:10:00.000Z	528	437	91	P1 Poststr	180	592	82.76515152	64
2022-10-28T10:11:00.000Z	528	436	92	P1 Poststr	180	592	82.57575758	64
2022-10-28T10:12:00.000Z	528	435	93	P1 Poststr	180	592	82.38636364	64
2022-10-28T10:13:00.000Z	528	439	89	P1 Poststr	180	592	83.14393939	64
2022-10-28T10:14:00.000Z	528	440	88	P1 Poststr	180	592	83.33333333	64
2022-10-28T10:15:00.000Z	528	439	89	P1 Poststr	180	592	83.14393939	64
2022-10-28T10:16:00.000Z	528	438	90	P1 Poststr	180	592	82.95454546	64
2022-10-28T10:17:00.000Z	528	437	91	P1 Poststr	180	592	82.76515152	64
2022-10-28T10:18:00.000Z	528	437	91	P1 Poststr	180	592	82.76515152	64
2022-10-28T10:19:00.000Z	528	438	90	P1 Poststr	180	592	82.95454546	64
2022-10-28T10:20:00.000Z	528	439	89	P1 Poststr	180	592	83.14393939	64
2022-10-28T10:21:00.000Z	528	441	87	P1 Poststr	180	592	83.52272727	64
2022-10-28T10:22:00.000Z	528	443	85	P1 Poststr	180	592	83.90151515	64
2022-10-28T10:23:00.000Z	528	444	84	P1 Poststr	180	592	84.09090909	64
2022-10-28T10:24:00.000Z	528	446	82	P1 Poststr	180	592	84.46969697	64
2022-10-28T10:25:00.000Z	528	445	83	P1 Poststr	180	592	84.28030303	64
2022-10-28T10:26:00.000Z	528	442	86	P1 Poststr	180	592	83.71212121	64
2022-10-28T10:27:00.000Z	528	444	84	P1 Poststr	180	592	84.09090909	64
2022-10-28T10:28:00.000Z	528	444	84	P1 Poststr	180	592	84.09090909	64
2022-10-28T10:29:00.000Z	528	444	84	P1 Poststr	180	592	84.09090909	64
2022-10-28T10:30:00.000Z	528	445	83	P1 Poststr	180	592	84.28030303	64
2022-10-28T10:31:00.000Z	528	447	81	P1 Poststr	180	592	84.65909091	64
2022-10-28T10:32:00.000Z	528	447	81	P1 Poststr	180	592	84.65909091	64
2022-10-28T10:33:00.000Z	528	449	79	P1 Poststr	180	592	85.03787879	64
2022-10-28T10:34:00.000Z	528	451	77	P1 Poststr	180	592	85.41666667	64
2022-10-28T10:35:00.000Z	528	450	78	P1 Poststr	180	592	85.22727273	64
2022-10-28T10:36:00.000Z	528	448	80	P1 Poststr	180	592	84.84848485	64
2022-10-28T10:37:00.000Z	528	447	81	P1 Poststr	180	592	84.65909091	64

Data Issues Found : Historical Data

observationDateTime	totalAvailableSpotNumber	occupiedSpotNumber	availableSpotNumber	name	maximumAllowedHeight	totalSpotNumber	aggregateRating	outOfServiceSlotNumber
2022-10-27T15:10:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:10:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:11:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:12:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:13:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:14:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:15:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:16:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:17:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:18:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:19:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:20:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:21:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:22:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:23:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:24:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:25:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:26:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:27:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:28:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:29:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:30:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:31:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:32:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:33:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:34:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:35:00.000Z	140	140	0	P3 Crown	200	130	100	-10
2022-10-28T10:36:00.000Z	140	140	0	P3 Crown	200	130	100	-10

Q&A

Data Freshness

"How frequently is this dataset updated, and how can we verify its timeliness for our analysis?"

Data Documentation

"Is there metadata or a data dictionary explaining field definitions?"

Next Steps

"Given the data quality, should we prioritize cleaning (e.g., handling missing values, standardizing formats) or explore alternative sources?"



THANK YOU